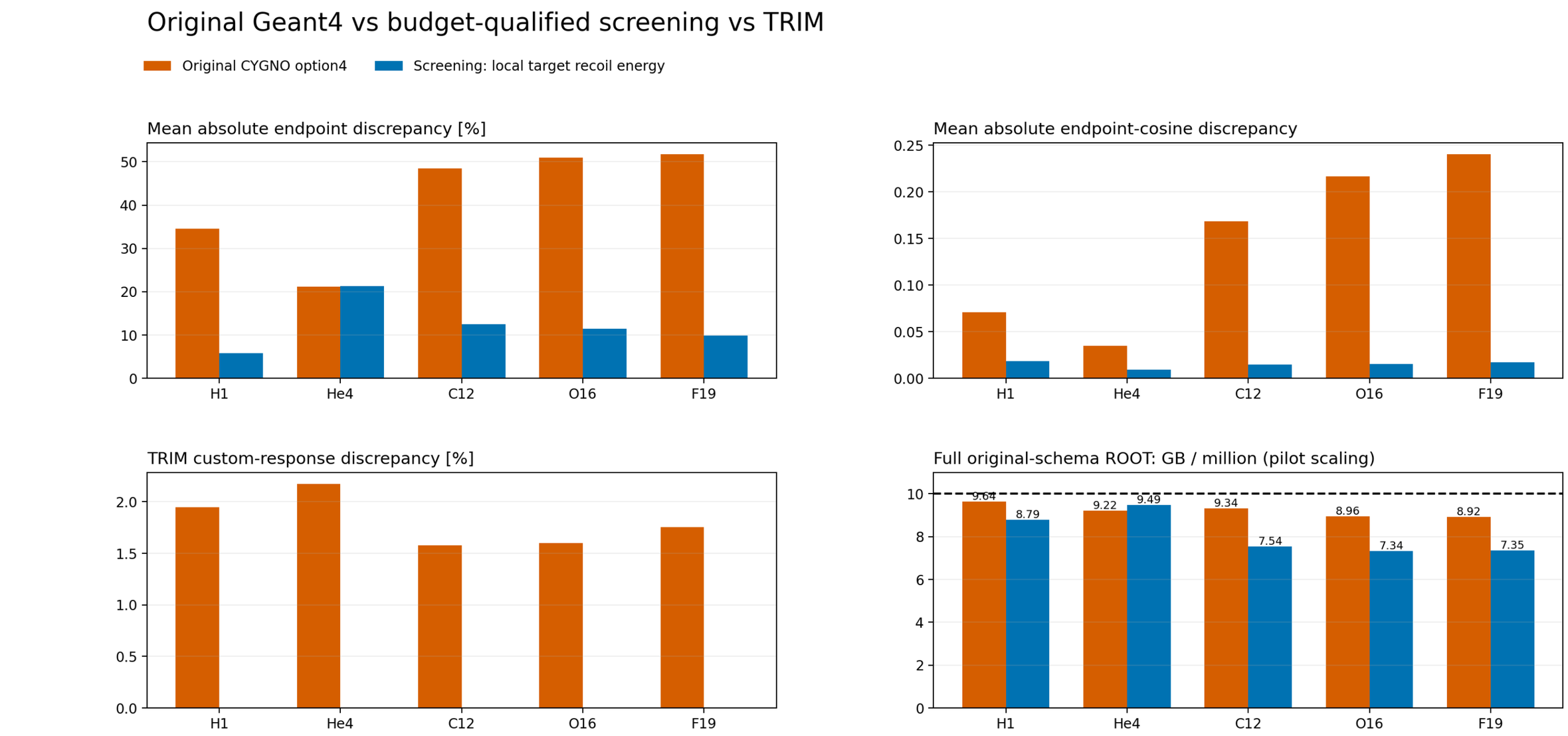
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Comparison overview



Fresh matched 10,000 events/ion/model: 100,000 events total. All original ROOT branches/types and one record per native depositing step are retained. New files project to 7.34-9.49 GB/million; these are 10,000-event measurements, not measured million-event files. The model deposits target recoil energy locally. Its close custom-QF agreement is imposed by the existing TRIM response integral, not an independent microscopic prediction. H/C/O/F ranges improve; He endpoint agreement is essentially unchanged.

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Models, sample and output contract

Original: g4v11.3_cygno04, commit 3f53d24. Screening implementation: 0151459. Both are tested with Geant4 11.4.0 and ROOT 6.36.08. The branch naming convention does not constitute a Geant4 11.3 test.

Each model has 10,000 new H1, He4, C12, O16 and F19 events. Initial energies are log-uniform from 0.25 to 1000 keV total, with isotropic directions at (0,0,245) mm in the existing 900 mbar, 293.15 K He/CF4 gas. Source arrays match bitwise between models. The original control is built from an immutable archive of its commit.

The ROOT writer, event action, sensitive detector and primary generator equal the original branch. Every branch name/type matches, and all six scalar energy channels match their vector sums. A bare-production/observed-build smoke test gives bitwise-identical ROOT arrays. No compact schema, merging, quantization, special reader, hit thinning or compression change is used.

Both retain option4 electronic ionisation and the existing other physics constructors. Screening replaces continuous nuclearStopping and low-energy ion multiple/Coulomb scattering by ZBL screened elastic collisions. Target recoil energy is deposited locally: the earlier fully tracked cascade is not retained. This is the central affordability approximation.

The transfer parameter used to construct cross-section/MFP tables is clamp(0.001*E, 0.1 eV, 100 eV), for total projectile energy E. The constructor low-energy floor is 0.1 eV, the terminal cutoff 1 eV/u, and the existing charged tracking threshold 25 eV. Macro cap factors are H/C/O/F x4 and He x8; the sqrt(E/40 keV) law remains.

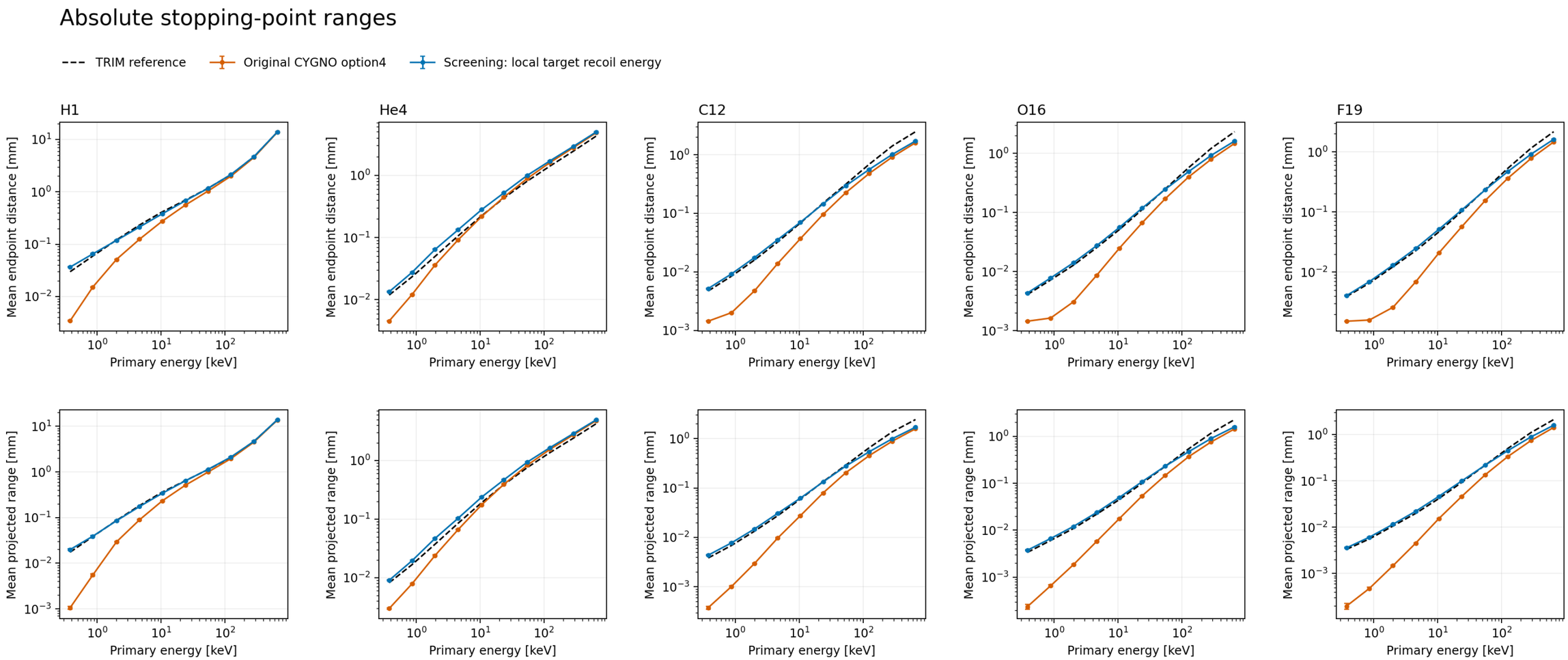
Quenching integrates the same hard-coded TRIM derivative: R(Epre)-R(max(0,Epre-local_deposit)), with R(E)=E*QF_TRIM(E). The earlier fitted event-gain table is removed. For contained primary-only local deposition the increments telescope to R(E0); matching this curve is by construction, with no new QF fit.

Ten logarithmic energy bins are used. Ratios average individual event/TRIM(E) values and scores equally weight the bins. Mean MC errors use event SEM. Range ratio errors also propagate shared TRIM node uncertainties; interpolation/model systematics are excluded. Backward fractions use Wilson 95% intervals.

The frozen TRIM range cohort has 195 energy/species points, 897 batches and 89,700 ions. TRIM electronic energy includes excitation and is not measured collected charge. Stopping-point spread curves include intrinsic spread plus the energy mixture within each bin. No per-event TRIM 3D charge cloud or intrinsic yield-fluctuation reference is available.

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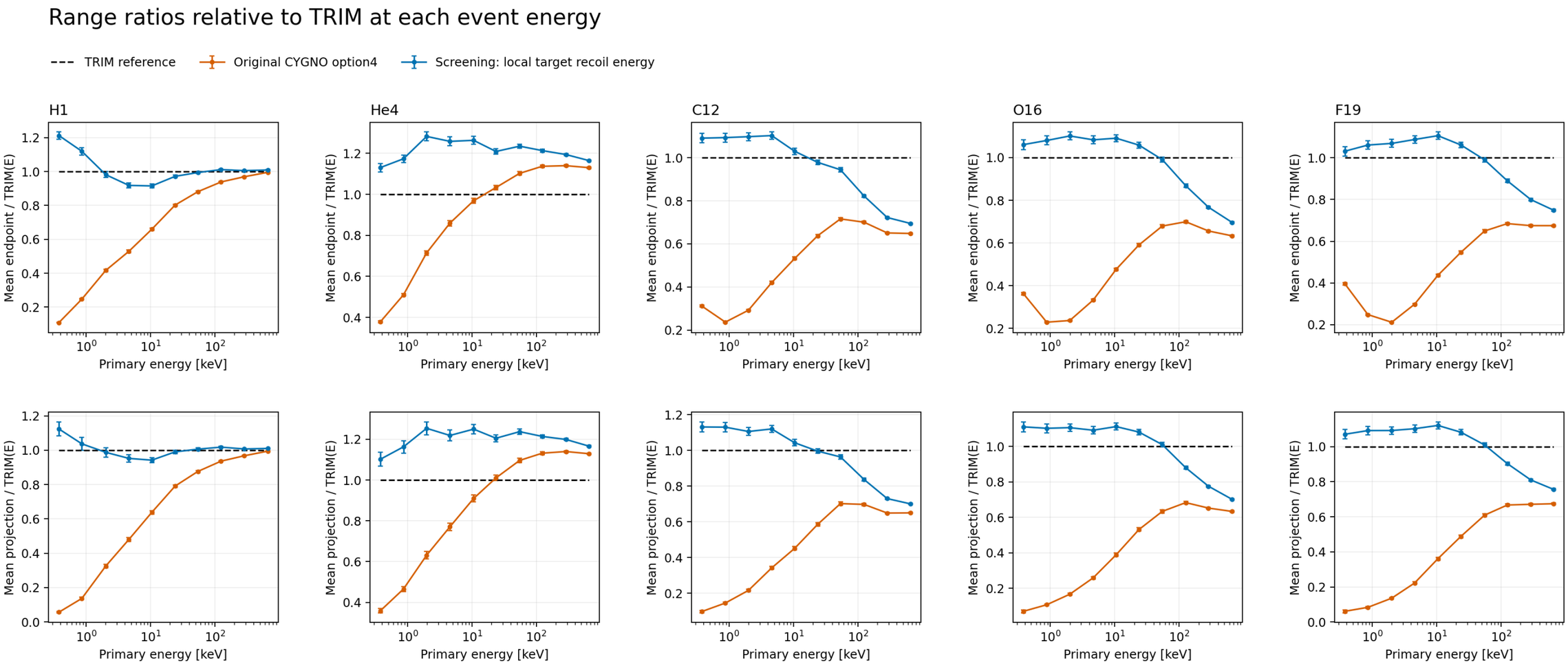
Absolute ranges



All primary tracks stop in sensitive gas. Endpoint is the source-to-stop distance; projected range is its component along the true initial direction. Points are averages in matched energy bins and bars are MC SEM. The dashed reference is the TRIM mean averaged over the same actual event energies. Geant4 true curved path is not equated with the sampled TRIM polyline.

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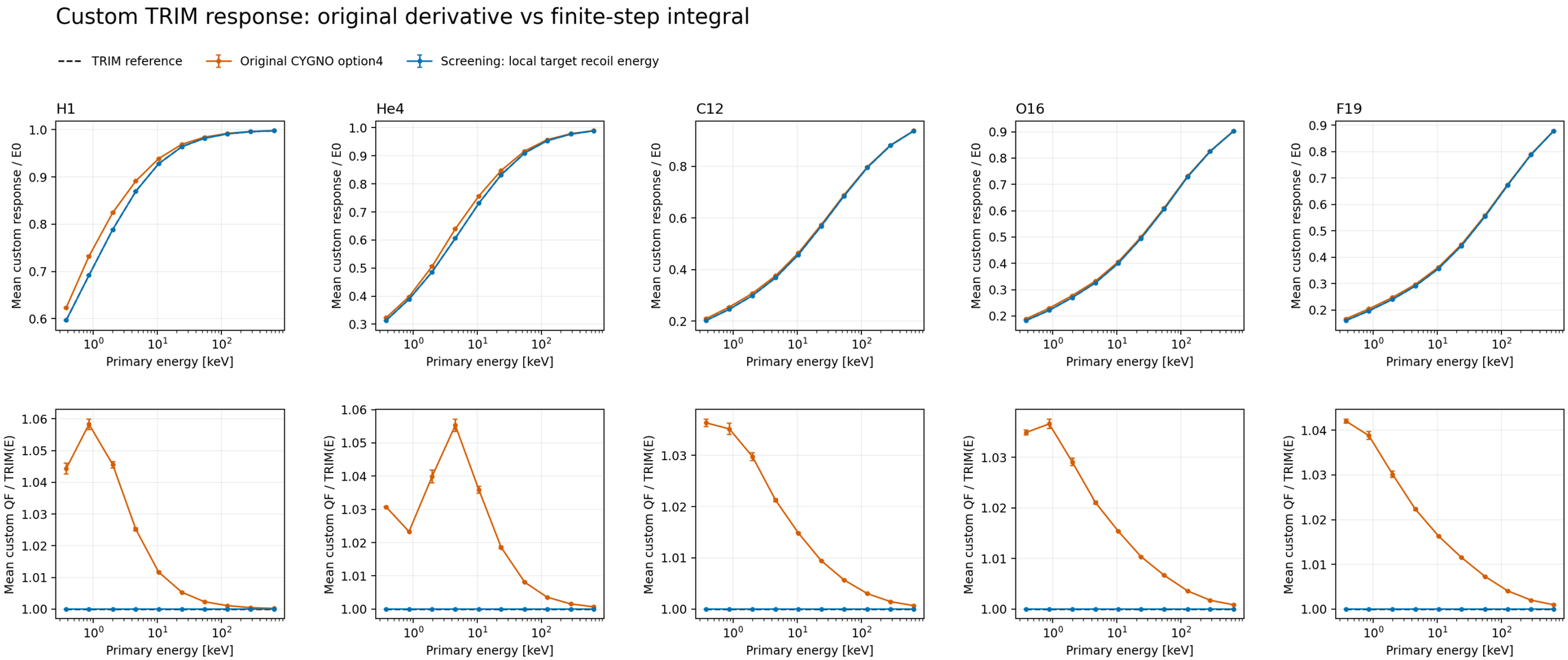
Range discrepancies



Ratios are means of individual energy-normalized observations, not ratios of binned means. Errors combine MC and conditional TRIM mean uncertainty, including shared interpolation-node correlations. Reference/model systematic uncertainty is excluded. H/C/O/F improve substantially on average; He endpoint agreement remains comparable. Residual discrepancies at particular energies remain visible.

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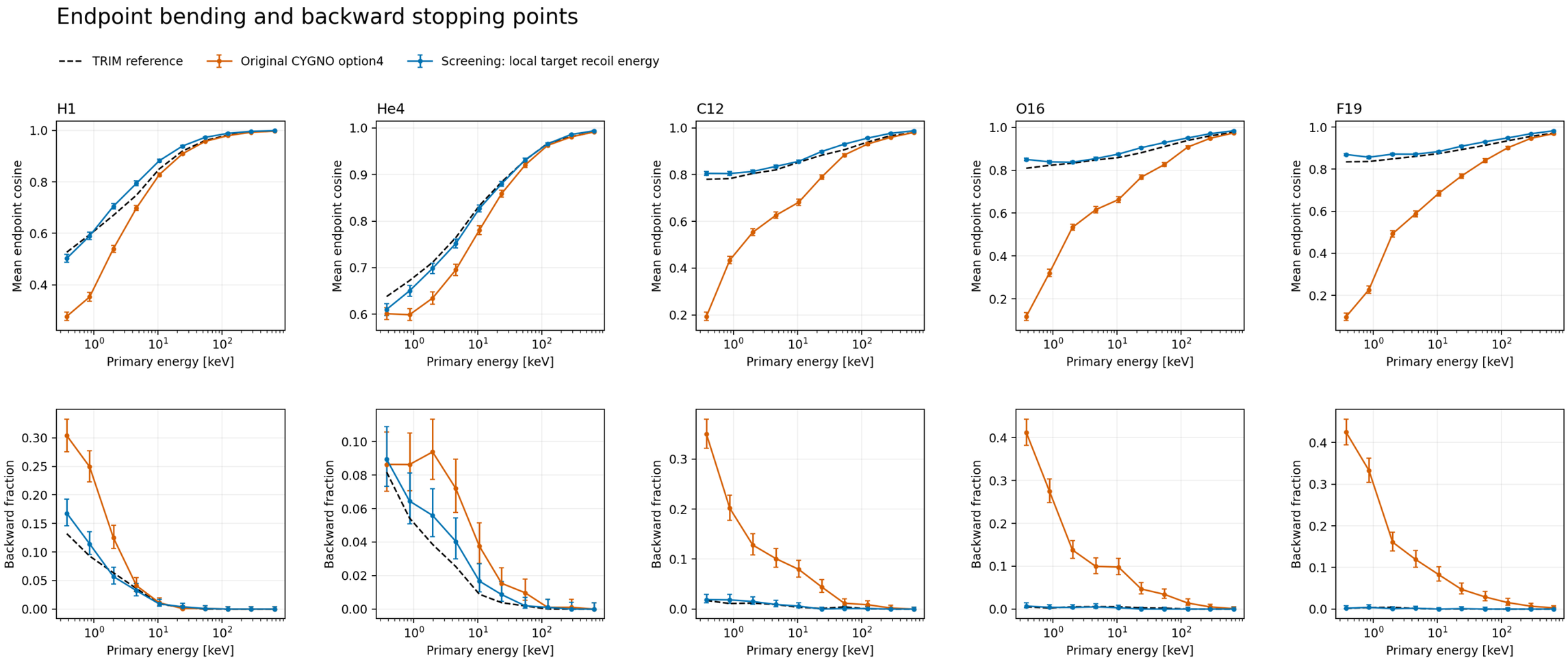
Quenching: construction and numerical check



The new custom response integrates the existing TRIM-derived R(E), so near-unity QF ratios follow by construction for primary-only contained recoils. This is not evidence of an independently predicted microscopic QF. Bars are MC SEM only; frozen-response model uncertainty is excluded. Raw deposited energy and native Geant4 ionising/NIEL-related channels are not rescaled. Mean collected-charge calibration and intrinsic Fano statistics remain separate.

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Direction preservation

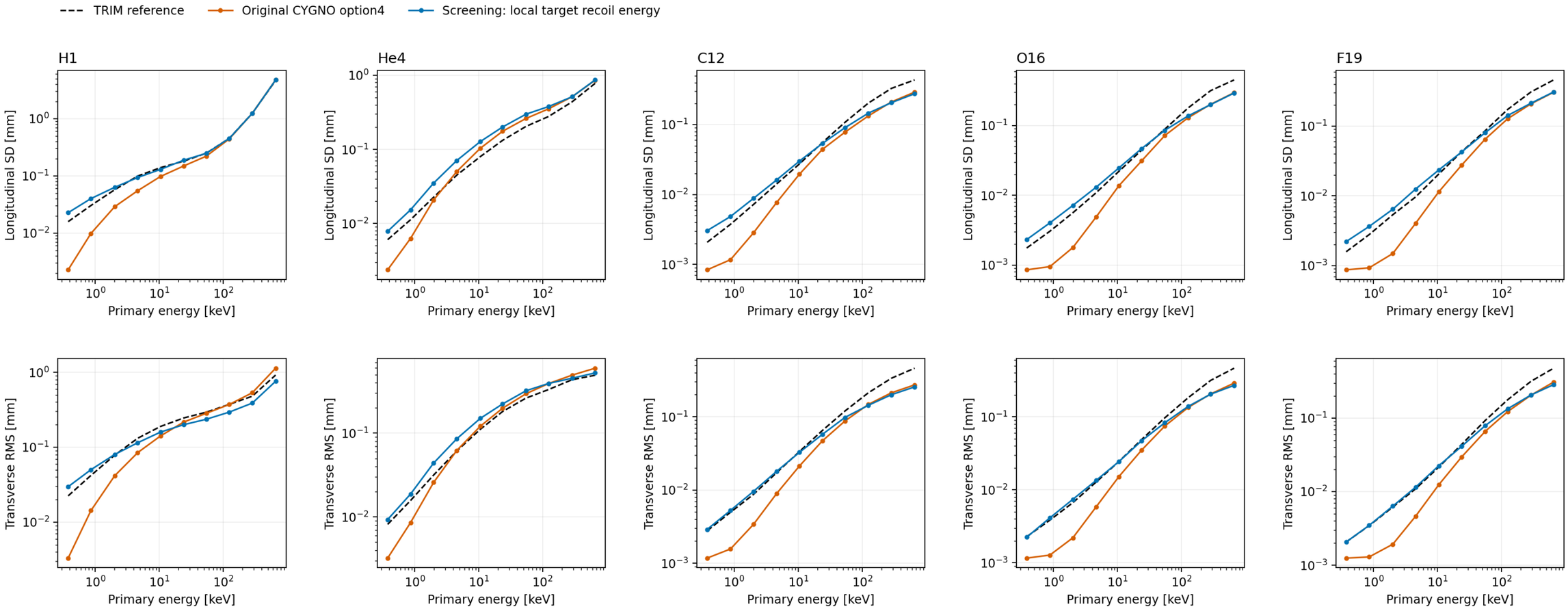


Endpoint cosine concerns initial momentum direction versus stopping-point displacement, not final momentum or reconstructed charge direction. Backward means negative projected displacement. Cosine bars are MC SEM; backward bars are Wilson 95% intervals. TRIM reference curves show means. Zero backward observations do not establish zero probability.

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Straggling

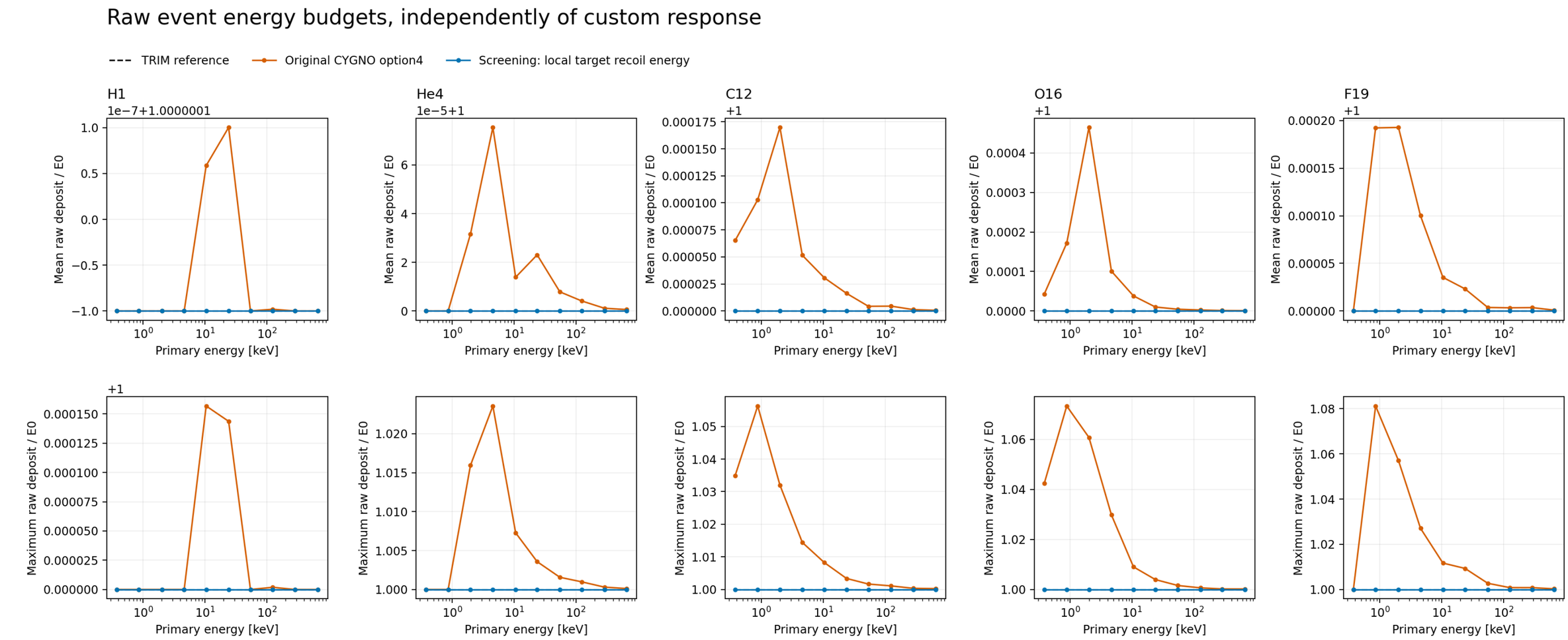
Stopping-point spread in the actual energy mixture



TRIM mixture longitudinal variance includes both mean per-energy variance and the variance of the energy-dependent projected mean. The transverse reference uses mean squared transverse displacement. References are reconstructed from frozen cohort means, SEMs and sizes. These spread curves have no uncertainty bands here; agreement of a mean range does not validate the full track morphology.

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Raw deposited energy

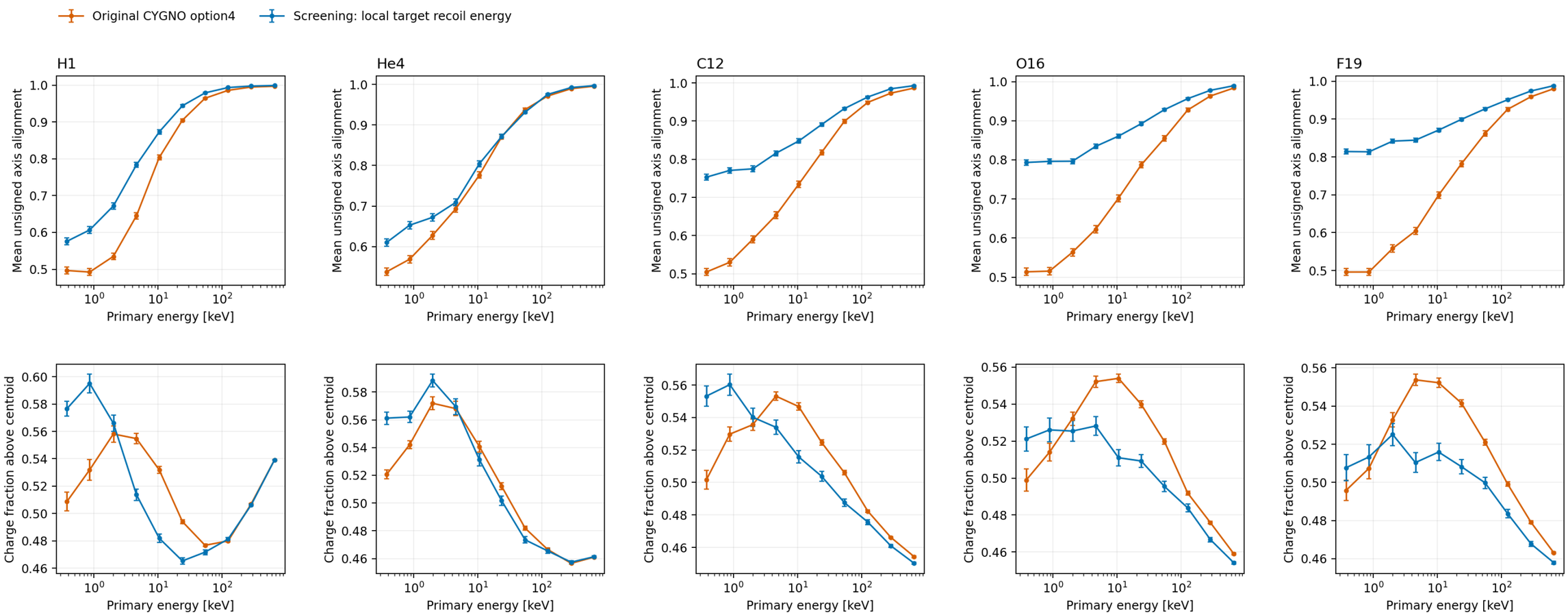


Unity is an energy-budget line, not a TRIM raw-deposit curve. All scored tracks are contained. Original option4 has 235/50,000 events above E0*(1+0.0001), with a largest excess of 8.12%; screening has zero such events. No event is removed from this comparison and no raw-energy normalization is applied. Maxima are sample maxima without confidence bars.

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Charge axis and two-half charge asymmetry

Recorded charge geometry: model comparison, no TRIM cloud reference

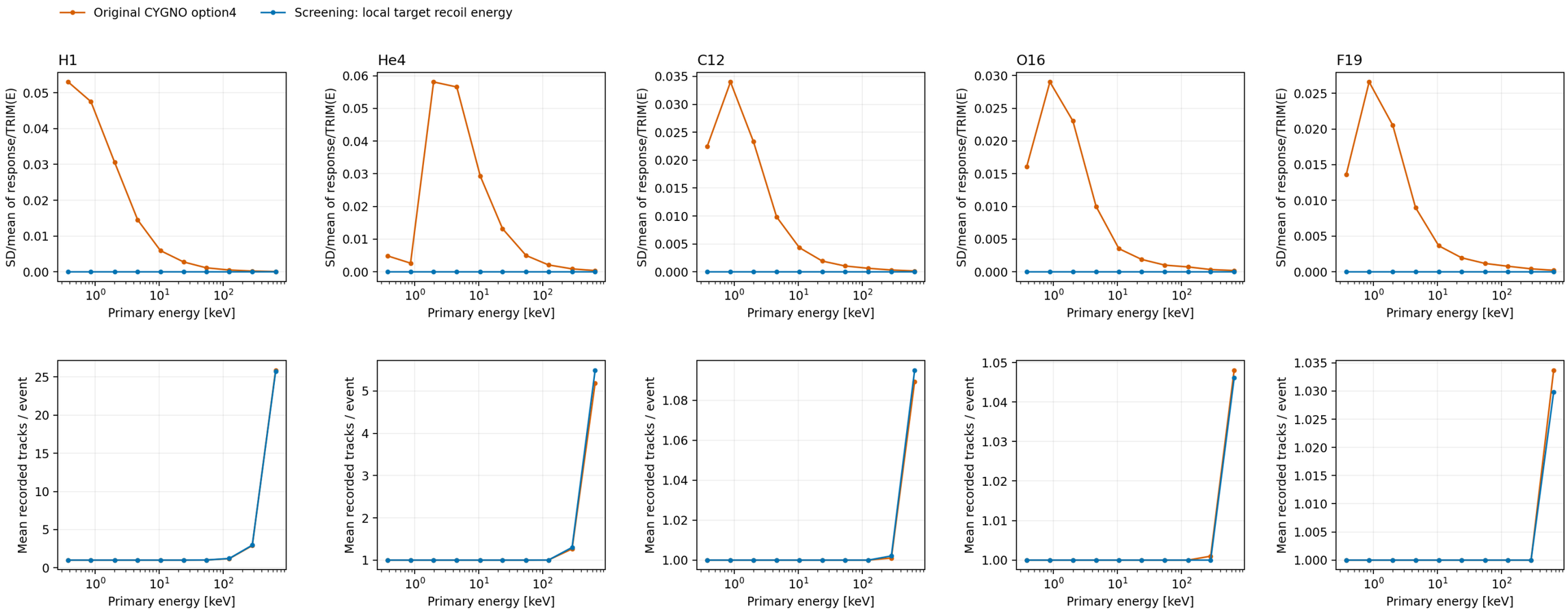


Weights are each model's custom QF hit energies. Axis alignment is the absolute dot product of the largest charge-covariance eigenvector and the initial direction. The two-half fraction splits projected charge at its centroid along the initial direction. These are model comparisons only: a per-event TRIM 3D charge reference is not available. Local target-energy deposition omits recoil-branch spatial correlations; these curves do not establish detector angular resolution or head-tail efficiency.

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Response fluctuations and track content

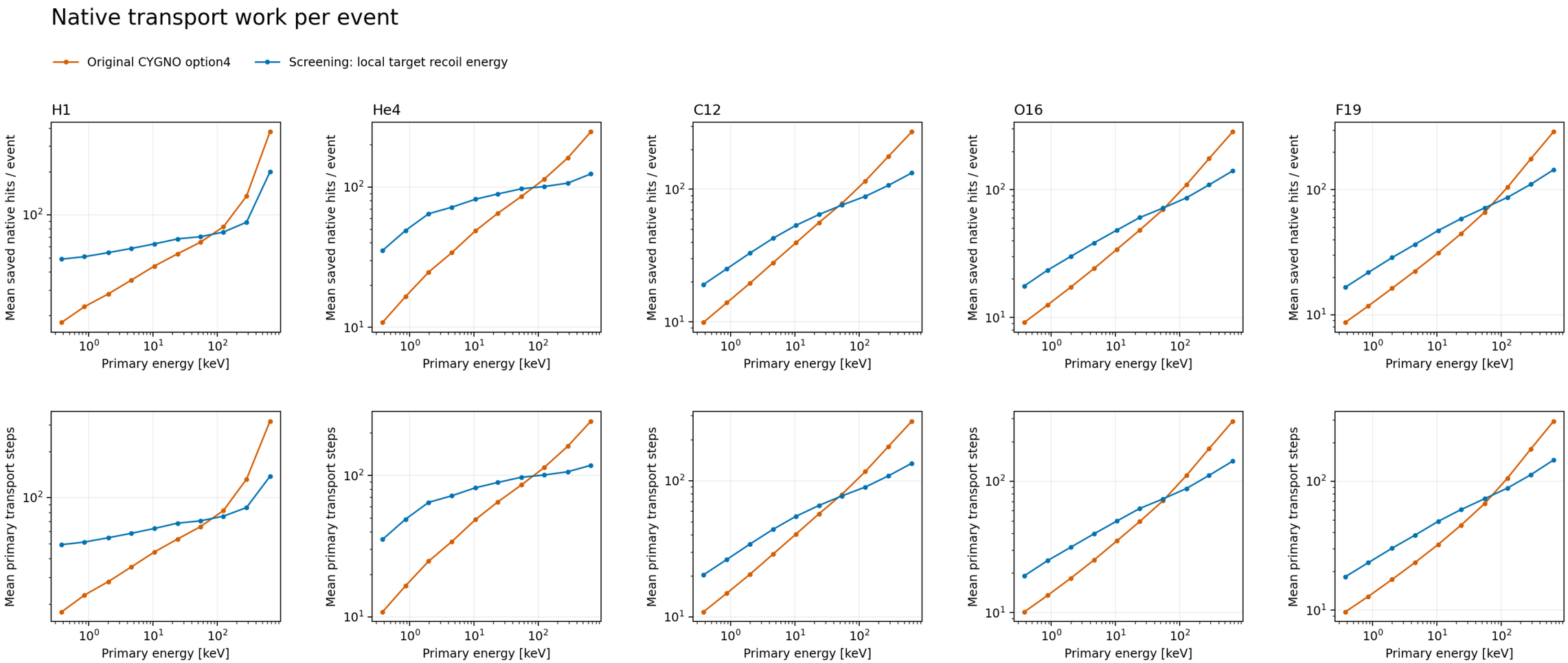
Response variation after removing energy-bin trend



Normalizing event response by R_TRIM(E0) removes the trend across the broad energy bin. New variation is nearly zero because the finite-step increments telescope; remaining effects include separate electronic secondaries. This is not a microscopic yield-variance or Fano prediction. Explicit target-recoil tracks are disabled; other registered processes may still generate secondary tracks. No TRIM fluctuation reference is invented.

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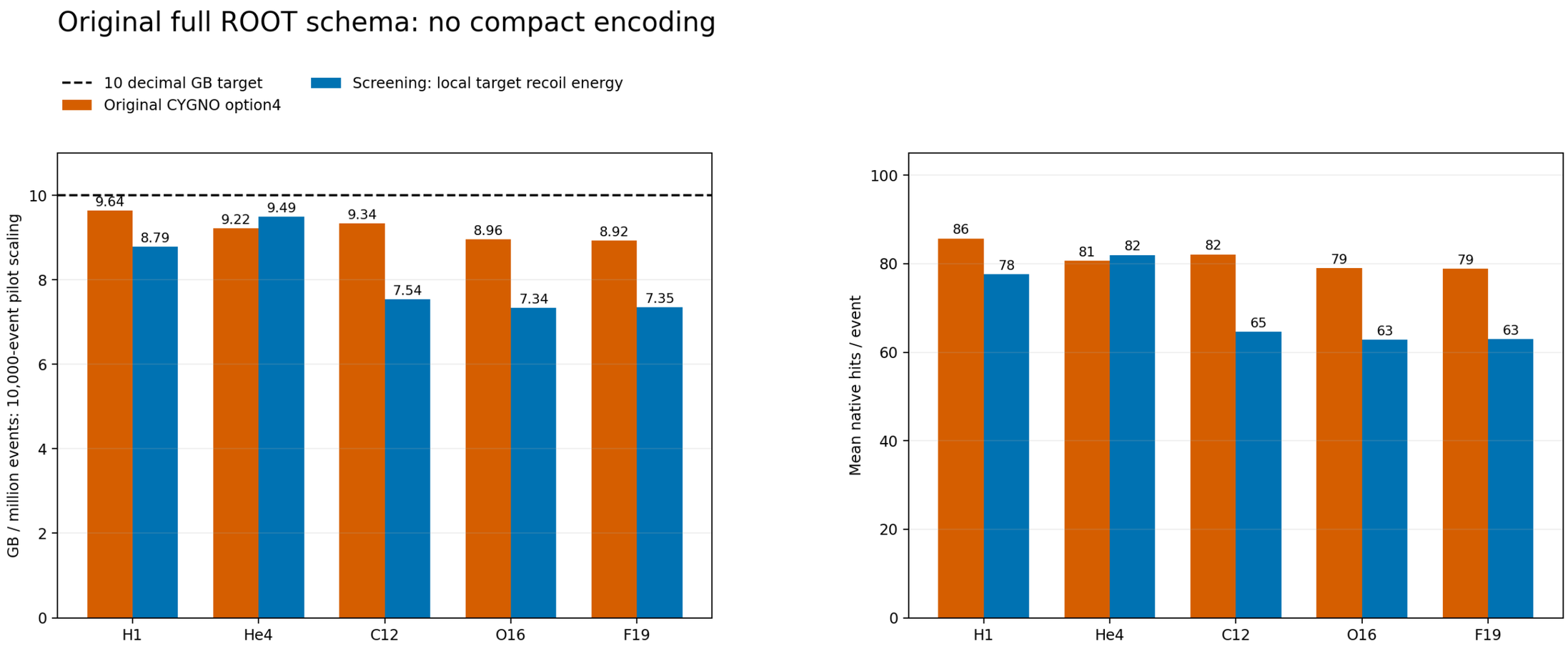
Native step counts



One saved hit remains one native depositing step. Reduction comes from transport configuration, not saved-hit averaging or omission. Step/track counts are work indicators; screened collision calculations differ in cost from option4 MSC. Pilot wall times were not collected as an isolated matched CPU/peak-memory benchmark, so no wall-time speedup is claimed.

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Full-format storage budget



Actual 10,000-event new files contain 87.86, 94.94, 75.45, 73.35 and 73.54 MB for H/He/C/O/F. Scaled projections are 8.786, 9.494, 7.545, 7.335 and 7.354 decimal GB/million. All branch names/types and energy-vector sums were checked. He has about 5% margin. These are pilot extrapolations for the stated spectrum/caps, not measured million-event files or a bound for arbitrary high-energy samples.

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Interpretation and remaining qualification

The budget is reached while preserving the original file layout and one-hit-per-native-step contract. The actual tested implementation is a primary-ion screened transport model with local target-recoil energy deposition; it is not the earlier fully tracked cascade hidden inside smaller files.

Matched-source endpoint mean absolute discrepancies, equally weighting ten energy bins, are original/new: H 34.52/5.77%, He 21.10/21.23%, C 48.52/12.52%, O 51.00/11.51%, F 51.80/9.88%. He endpoint agreement is essentially unchanged. In the highest bin (about 0.66 MeV geometric mean), new endpoint/TRIM ratios are H 1.009, He 1.164, C 0.695, O 0.696 and F 0.749: C/O remain approximately 30% short and F 25% short. Average improvement does not establish agreement at every energy.

Custom QF agreement is a check of the imported mean-response function and its finite-step integration. The original derivative approximation has step-size quadrature bias; the new integral removes it. Intrinsic detector yield fluctuations, excitation-to-charge conversion, Fano statistics and measured charge/light calibration are not established by this agreement.

Charge geometry requires additional detector-response validation. Target-recoil branches and their spatial correlations are locally approximated; fine primary endpoint agreement alone cannot validate the full recoil cascade, reconstructed angle or head-tail efficiency. The original model is not treated as a detector-truth reference.

The documented size target applies to each ion's 0.25-1000 keV log-uniform spectrum with the branch macros. He has the smallest projected margin. A different spectrum, source geometry, gas or step cap requires renewed qualification. An actual million-event file has not been measured here.

The comparison CSV, technical note and validation metadata accompany these PDFs. Local ROOT and primary-ledger outputs, executable hashes and source hashes are retained under tests/nuclear_recoils/screening_development/budget_qualification. Monte Carlo output files and exploratory code are not imported into the minimal production patch.